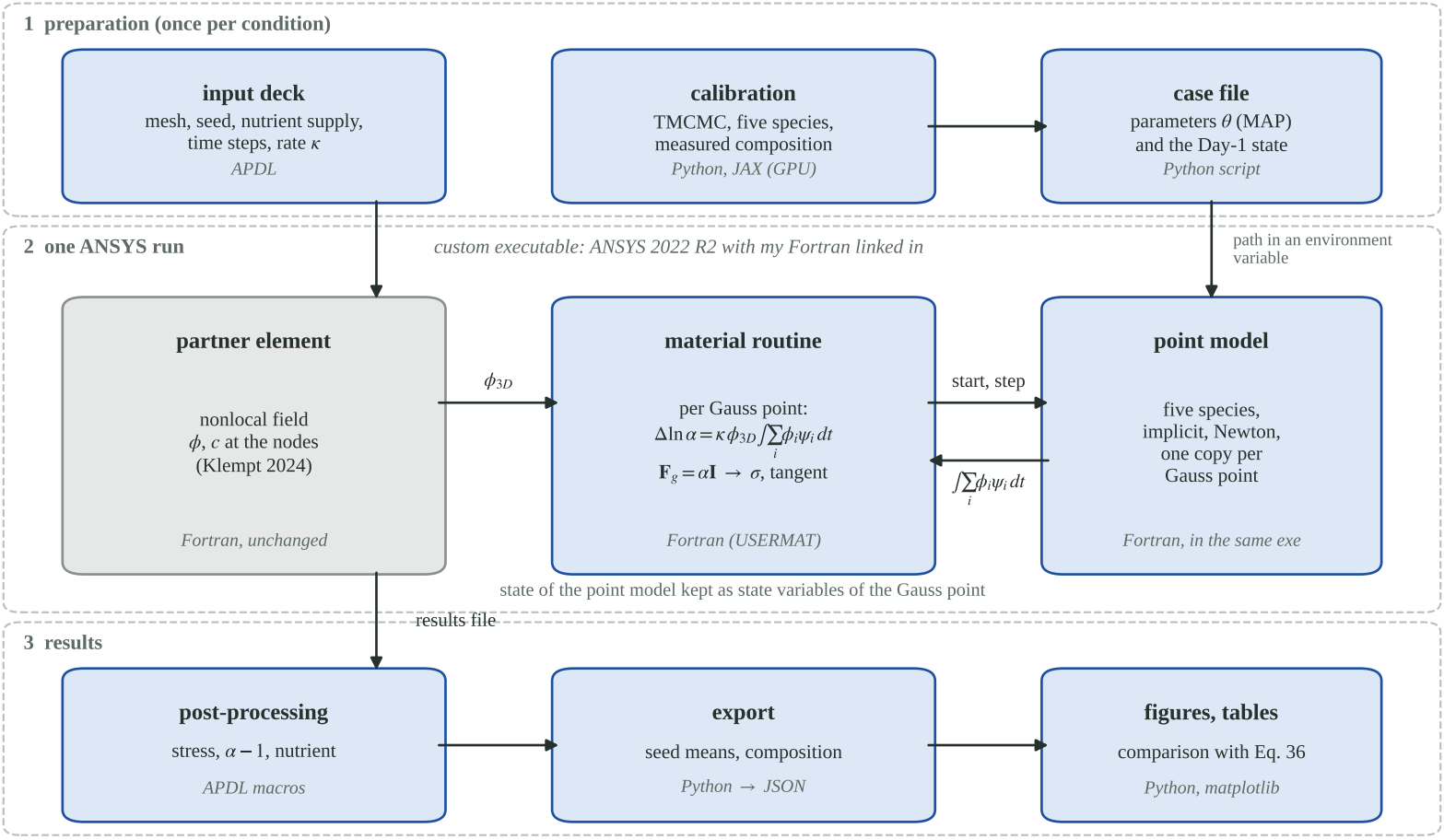

1 preparation (once per condition)
input deck
mesh, seed, nutrient supply, time steps, rate κ
APDL
calibration
TMCMC, five species, measured composition
Python, JAX (GPU)
case file
parameters θ (MAP) and the Day-1 state
Python script
2 one ANSYS run
custom executable: ANSYS 2022 R2 with my Fortran linked in
path in an environment variable
partner element
nonlocal field ϕ, c at the nodes (Klempt 2024)
Fortran, unchanged
ϕ3D
material routine
per Gauss point:
Δln α = κ ϕ3D ∫ Σi ϕi ψi dt
Fg = αI → σ, tangent
Fortran (USERMAT)
start, step
∫ Σi ϕi ψi dt
point model
five species, implicit, Newton, one copy per Gauss point
Fortran, in the same exe
state of the point model kept as state variables of the Gauss point
results file
3 results
post-processing
stress, α − 1, nutrient
APDL macros
export
seed means, composition
Python → JSON
figures, tables
comparison with Eq. 36
Python, matplotlib
Build: Intel Fortran 2025 + Visual Studio, one link script. Runs: PowerShell, one ANSYS run at a time, detached from the session.
added in this work
used as it is